

Recommended Assessment

Joysticks and Potentiometers

Exploring Potentiometer Inputs

1. Write an equation for the voltage V_p across the potentiometer with resistance R_p . What is the V_p when $R_p = 0\Omega$? What is the V_p when $R_p = 100k\Omega$?
2. What happens to the potentiometer signal when you rotate the potentiometer knob?
3. How does the potentiometer signal change when you instead connect the left and middle terminals of the potentiometer to the device? What is the relationship between the angular position of the potentiometer knob and the measured voltage based on your observations? How did you verify this using the datasheet?
4. When the potentiometer is rotated to its limits in each direction, do they match the voltages at the minimum and maximum resistances given by the equation from step 2?
5. When the potentiometer slowly rotated between its limits at a consistent speed, does the estimated angle change consistently with the potentiometer rotation? What happens when the potentiometer is rotated near its limits?
6. What is the potentiometer's angular range? Given its angular range, how would you use the measured voltage to obtain the angular position of the potentiometer shaft? Write the equation to calculate the shaft angle, θ , given the measured voltage, V_p .

Exploring Joystick Inputs

7. How do the Joystick X and Joystick Y signals change as you move the joystick in each direction? What happens when you click the joystick?
8. Compare the joystick to the potentiometer. How do they differ in the way that you move them? How does the signal differ? How are they the same?
9. What happens when the joystick quickly bounces back to the center compared to being slowly moved back to the center? Joysticks are often used to control movement in applications such as video game or aircraft controllers. What should the signals ideally read when the joystick is in the center and why is this important?
10. How would you obtain the direction of the joystick using the Joystick X and Joystick Y readings?

Potentiometer Cheat Sheet

What is it?	
What does it measure directly?	
What can you measure indirectly?	
Where are they used?	
Pros	
Cons	
Notes/ Important things to remember	